Indian Disease Data Analysis

1. Project Overview

This project analyses an Indian diseases data-set to understand disease patterns, patient health conditions, and recovery trends. The objective is to clean and process patient data using Python and store the refined data-set in a MySQL database. The analysis helps identify relationships between diseases, commodities, recovery time, and lifestyle factors to support better Health-care insights.

2. data-set Summary

Rows: **10,000**

Columns: **12**

- **Key Features:**

Patient Demographics (Age Group, Blood Group, BMI)

Disease Information (Disease Name, Symptoms)

Health Conditions (Co-morbidity, Alcohol Use)

Medical Details (Diagnosis Date, Recovery Days, Cause of Death)

- **Missing Data:**
  Missing values in **Co-morbidity** and **alcohol_use** were replaced with "never".
  Missing values in **cause_of_death** were filled with "Unknown".
  Missing values in **recovery_days** were handled using the **median value grouped by disease name**.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

● **Data Loading:** Imported the Indian diseases data-set using **pandas** into a Data-frame for further analysis.

● **Initial Exploration:** Used **data.info()** to examine the data-set structure and **data.describe()** to generate summary statistics for numerical columns.

```
                age          year           bmi  days_hospitalized  treatment_cost_inr    death_flag  recovery_days
count  20000.000000  20000.000000  20000.000000       20000.000000        2.000000e+04  20000.000000   13693.000000
mean      44.061050   2021.498000     23.485495           9.771150        1.079405e+05      0.091100      33.205141
std       25.165393      2.294179      4.666045          12.078476        4.940645e+05      0.287758      26.427051
min        0.000000   2018.000000     14.000000           0.000000        1.200000e+01      0.000000       3.000000
25%       23.000000   2019.000000     19.700000           2.000000        1.421000e+03      0.000000      14.000000
50%       44.000000   2022.000000     23.200000           6.000000        6.604500e+03      0.000000      24.000000
75%       63.000000   2023.000000     27.000000          14.000000        3.933375e+04      0.000000      47.000000
max       95.000000   2025.000000     40.000000          60.000000        1.537582e+07      1.000000     100.000000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
```

```
Data columns (total 32 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   patient_id          20000 non-null  object
 1   age                 20000 non-null  int64
 2   age_group           20000 non-null  object
 3   gender              20000 non-null  object
 4   state               20000 non-null  object
 5   city                20000 non-null  object
 6   region              20000 non-null  object
 7   urban_rural         20000 non-null  object
 8   disease_name        20000 non-null  object
 9   disease_category    20000 non-null  object
 10  severity            20000 non-null  object
 11  diagnosis_date      20000 non-null  object
 12  year                20000 non-null  int64
 13  month               20000 non-null  object
 14  season              20000 non-null  object
 15  symptoms            20000 non-null  object
 16  comorbidity         13401 non-null  object
 17  smoking_status      20000 non-null  object
 18  alcohol_use         13373 non-null  object
 19  bmi                 20000 non-null  float64
 20  blood_group         20000 non-null  object
 21  income_level        20000 non-null  object
 22  insurance_status    20000 non-null  object
 23  treatment_type      20000 non-null  object
 24  hospital_type       20000 non-null  object
 25  days_hospitalized   20000 non-null  int64
 26  treatment_cost_inr  20000 non-null  float64
 27  outcome             20000 non-null  object
 28  death_flag          20000 non-null  int64
 29  cause_of_death      1822 non-null   object
 30  recovery_days       13693 non-null  float64
 31  follow_up_required  20000 non-null  object
dtypes: float64(3), int64(4), object(25)
memory usage: 4.9+ MB
```

● **Missing Data Handling:** Checked for null values across all columns. Missing values in co-morbidity and **alcohol_use** were filled with *"never"*. Missing values in **cause_of_death** were replaced with *"Unknown"*. Missing values in **recovery_days** were imputed using the **median value grouped by disease_name**.

● **Column Standardization:** Standardized categorical values in columns such as **age_group** by correcting inconsistent values like *"19-oct"* to *"0-20"* and *"80+"* to *"80-90"*.

● **Feature Processing:** Converted **diagnosis_date** column into proper date time **format** to enable time-based analysis.

● **Data Consistency Check:** Verified the data-set for duplicate records and checked categorical columns such as co-morbidity, **alcohol_use**, **blood_group**, and **symptoms** for consistent values.

● **Database Integration:** Connected the Python script to **MySQL** using SQLAlchemy and successfully loaded the cleaned DataFrame into the **indian_diseases_data** database table for further SQL-based analysis.

Data Analysis using SQL (Business Transactions)

We performed structured analysis in MYSQL to answer key Health-care questions:

1. **Total Number of Patients** – Calculated the total number of patients registered in the data set.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_patients
20000

2. **Disease with Highest Cases** – Identified the disease with the highest number of reported cases.

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

disease_name	patient_count
Leptospirosis	664
Asthma	660
Hypertension	654
Gastroenteritis	654
Oral Cancer	652

3. **Average Recovery Time by Disease** – Calculated the average recovery time for each disease.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

disease_name	avg_recovery_days
Japanese Encephalitis	48.26
Stroke	44.42
Lung Cancer	44.05
Cholera	42.37
Tuberculosis	41.3
Heart Disease	38.84
Kala-Azar	38.34
Oral Cancer	36.14
Breast Cancer	35.09
Pneumonia	34.05
Dengue	32.65
Leptospirosis	32.41
Swine Flu (H1N1)	31.91
Malaria	29.6

4. **Total Recorded Deaths** – Determined the total number of deaths recorded in the data set.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_deaths
1822

5. **Highest Mortality Rate Disease** – Identified the disease with the highest mortality rate.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

disease_name	mortality_rate
Stroke	31.69130
Heart Disease	25.08418
Lung Cancer	23.27150
Oral Cancer	22.69939
Chronic Kidney Disease	22.38095
Breast Cancer	19.87076
Hypertension	17.12538
Diabetes (Type 2)	14.53958
COPD	14.19458
Pneumonia	10.93502
Asthma	9.09091
Swine Flu (H1N1)	8.95522
Tuberculosis	8.38926
Japanese Encephalitis	8.16000

6. **Gender Distribution of Patients** – Compared the number of male and female patients.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

gender	total
Male	9926
Female	9873
Other	201

7. **Monthly Case Trends** – Calculated the number of cases reported each month based on the diagnosis date.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

month	total_cases
2018-01	241
2018-02	218
2018-03	155
2018-04	184
2018-05	170
2018-06	162
2018-07	242
2018-08	264
2018-09	234
2018-10	219
2018-11	225
2018-12	233

8. **Alcohol Use vs Recovery Time** – Analyzed whether alcohol users have different average recovery times compared to non-alcohol users.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

alcohol_use	avg_recovery
never	30.91
Regular	31.87
Occasional	30.55
Heavy	31.02

9. **Co-morbidity Risk Analysis** – Checked whether patients with commodities have a higher risk of death compared to those without commodities.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

comorbidity	deaths	total_patients
never	527	6599
Diabetes	232	2242
Heart Disease	223	2217
Multiple	222	2219
Kidney Disease	220	2314
Hypertension	206	2173
Obesity	192	2236

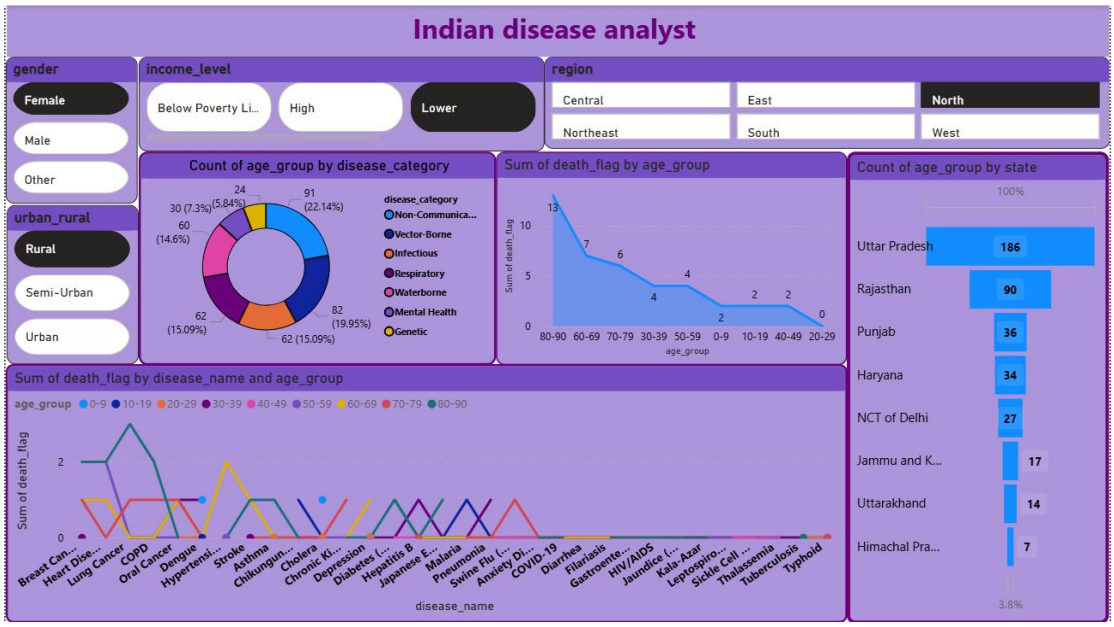
10. **Longest Recovery Cases** – Identified the five patients with the longest recovery duration.

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

patient_id	disease_name	recovery_days
IND-00018885	Japanese Encephalitis	100
IND-00019192	Leptospirosis	100
IND-00018915	Japanese Encephalitis	100
IND-00019821	Swine Flu (H1N1)	100
IND-00019565	Hepatitis B	100

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

● **Focus on High-Risk Diseases:**
Diseases with higher case counts or higher mortality rates should receive greater attention from health-care authorities for better prevention and treatment planning.

● **Early Detection Programs:**
Hospitals and Health-care organizations should promote regular health check-ups and early diagnosis programs to reduce the severity of diseases and improve recovery rates.

● **Lifestyle Awareness Campaigns:**
Public awareness campaigns should be conducted to educate people about the impact of lifestyle factors such as alcohol consumption and unhealthy habits on disease recovery.

● **Better Monitoring of Patients with Commodities:**
Patients with existing health conditions should be closely monitored, as they may have a higher risk of complications and mortality.

● **Improve Health-care Resource Allocation:**
Health-care providers should allocate more resources, medical staff, and facilities to diseases that show longer recovery times or higher case volumes.

● **Use Data for Health-care Planning:**
Hospitals and health authorities should regularly analyze Health-care data to identify disease trends, monitor patient recovery patterns, and improve decision-making.